

Supplement to
*Generators and Relations for
Discrete Groups*

Hilbert Reconstruction and Lean Formalization

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Document Status

This volume is a research supplement accompanying the CGJteamLab formalization project. Its purpose is to investigate the generators-and-relations viewpoint of Coxeter and Moser inside a synthetic Hilbert geometry formalized in Lean.

The source book is H. S. M. Coxeter and W. O. J. Moser, *Generators and Relations for Discrete Groups*, second edition.

The present supplement does not reproduce the source book. Instead, it develops a parallel formal program:

Hilbert geometry $\longrightarrow$ geometric transformations $\longrightarrow$ generators $\longrightarrow$ words $\longrightarrow$ relations.

The Lean source is authoritative for theorem statements, assumptions, dependencies, and proof status. This document records the mathematical architecture, formal dependency structure, and relation to the Coxeter–Moser viewpoint.

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Research Program

The initial research program is deliberately narrow.

1. Reconstruct reflection in a line from the established Hilbert incidence, order, congruence, midpoint, and right-angle theory.
2. Promote the geometric reflection relation to an actual permutation of the point set.
3. Prove the first Coxeter relation

$$r_l^2 = 1.$$

4. Construct words in several reflection generators.
5. Study the pairwise relations

$$(r_l r_m)^p = 1$$

for intersecting axes.

6. Treat the parallel-axis case, corresponding to infinite dihedral behavior and translation.
7. Use selected propositions of Euclid's *Elements* as benchmark configurations for the generators-and-relations language.

The principal Coxeter–Moser source locations for the opening phase are Sections 1.1, 4.1, 4.3, and 9.1–9.3.

Chapter 1

Line Reflection as a Hilbert Generator

1.1 Purpose and Status

This document records the first completed component of the experimental Coxeter layer of CGJteamLab. The source module is

```
CGJteamLab/Coxeter/Reflection.lean
```

The module is deliberately separated from the flat sequence of Euclid Proposition files. It is not another proposition of the *Elements*; it is a transformation layer derived over the same Hilbert foundation.

The immediate objective is to obtain a genuine reflection generator

$$r_l : \text{Point} \longrightarrow \text{Point}$$

without introducing reflection as a new primitive geometric operation. The construction starts with a geometric relation, proves existence and uniqueness, and only then packages the relation as a point transformation.

The current file compiles cleanly. No new geometric axiom and no `sorry` is introduced locally.

1.2 Coxeter Motivation

Coxeter and Moser begin with generators and defining relations and later specialize to groups generated by reflections. In the reflection-group language, each basic reflection is involutory:

$$r_i^2 = 1,$$

while the geometry of pairs of reflecting hyperplanes is encoded by the periods of products

$$(r_i r_j)^{p_{ij}} = 1.$$

Their graphical notation records these pairwise periods by a Coxeter diagram.

The present module addresses only the first relation. It constructs a single Hilbert line reflection and proves, in literal operator form,

$$r_l(r_l(P)) = P.$$

Products of distinct reflections and relations of the form $(r_l r_m)^p = 1$ belong to the next layer and are intentionally absent from this file.

The relevant source locations in Coxeter–Moser, *Generators and Relations for Discrete Groups*, second edition, are Sections 1.1, 4.3, and 9.1–9.3.

1.3 Geometric Design

1.3.1 Why the construction begins as a relation

The Hilbert development does not take a global transformation space as a primitive object. The natural first notion is therefore a relation between a point P and its reflected point P' .

For a point off the axis l , reflection is characterized by a foot H satisfying

$$H \in l, \quad P - H - P', \quad PH \cong HP',$$

together with the condition that PH is perpendicular to the axis.

For a point on the axis, the reflected point is the point itself.

Thus the geometric core is

$$P' = P \text{ on } l \quad \text{or} \quad H \in l, H = \text{mid}(P, P'), PH \perp l.$$

1.3.2 ReflectionAxis

The current Hilbert API often represents a line geometrically by two distinct points known to lie on it. The reflection axis therefore stores both the carrier and witnesses sufficient for perpendicular constructions.

```
structure ReflectionAxis
  [HilbertIncidence Geo] where
  carrier : Geo.Line
  A : Geo.Point
  B : Geo.Point
  hAB : Not (A = B)
  hA : HilbertIncidence.OnLine A carrier
  hB : HilbertIncidence.OnLine B carrier
```

This is not additional geometry. It packages the nondegenerate line data already required by the existing Hilbert construction interface.

1.3.3 PerpendicularToAxis

Perpendicularity to the axis is represented using the established point-based right-angle language.

```

def PerpendicularToAxis
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (H P : Geo.Point) : Prop :=
And
  (HilbertIncidence.OnLine H axis.carrier)
  (And
    (Not (HilbertIncidence.OnLine P axis.carrier))
    (Exists fun R : Geo.Point =>
      And
        (HilbertIncidence.OnLine R axis.carrier)
        (And
          (Not (R = H))
          (HilbertRightAngle Geo R H P))))))

```

The witness R supplies a nondegenerate direction of the carrier. No primitive relation “two lines are perpendicular” is added.

1.3.4 IsLineReflection

The reflection relation has a fixed-point branch and an off-axis branch.

```

def IsLineReflection
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P P' : Geo.Point) : Prop :=
Or
  (And
    (HilbertIncidence.OnLine P axis.carrier)
    (P' = P))
  (And
    (Not (HilbertIncidence.OnLine P axis.carrier))
    (Exists fun H : Geo.Point =>
      And
        (PerpendicularToAxis Geo axis H P)
        (HilbertIsMidpoint Geo H P P'))))

```

The midpoint condition packages both strict betweenness $P - H - P'$ and equality of the two half-segments.

1.4 Existence of the Reflected Point

The first main theorem is

```

theorem line_reflection_exists
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P : Geo.Point) :

```

```
exists P' : Geo.Point,
  IsLineReflection Geo axis P P'
```

The proof splits on whether P lies on the axis.

If $P \in l$, choose $P' = P$.

If $P \notin l$, Euclid I.12 in its Hilbert reconstruction supplies the perpendicular foot H . The established theorem

```
hilbert_extend_segment_beyond
```

then extends PH beyond H to a point P' satisfying

$$P - H - P', \quad PH \cong HP'.$$

This is exactly the midpoint condition needed by `HilbertIsMidpoint`.

The conceptual dependency is therefore

perpendicular construction + equal segment extension $\implies$ existence of reflection.

1.5 Relational Involution

Before constructing a function, the development proves that the reflection relation is symmetric:

```
theorem line_reflection_symmetric
...
(hRef : IsLineReflection Geo axis P P') :
  IsLineReflection Geo axis P' P
```

For off-axis points, the same midpoint H is used in the reverse direction. The substantive geometric obligation is to transport the right-angle condition from the ray HP to the opposite ray HP' . This uses the already developed theory of right-angle transport and opposite extensions.

This theorem is the relational precursor of the Coxeter involution relation:

$$P R_l P' \implies P' R_l P.$$

1.6 Fixed Points

The module records the exact fixed-point set of the reflection:

```
theorem line_reflection_fixed_iff_on_axis
...
(P : Geo.Point) :
  IsLineReflection Geo axis P P <->
    HilbertIncidence.OnLine P axis.carrier
```

Thus

$$r_l(P) = P \iff P \in l$$

once the functional reflection has been constructed.

The companion theorem `line_reflection_moves_off_axis` states that an off-axis point cannot be fixed.

1.7 Right-Angle Carrier Transport

The uniqueness proof exposed a reusable gap in the permanent Hilbert API: a right angle should not depend on which non-vertex point is chosen on the same carrier of one arm.

Three local theorems were therefore established:

```
coxeter_right_angle_sameRay_first
coxeter_right_angle_opposite_first
coxeter_right_angle_collinear_first
```

The first transports a right angle along the same ray. The second reverses the first arm across the vertex. The third combines the two cases using betweenness trichotomy and gives the carrier-level statement required for perpendicular uniqueness.

This is a useful example of the formal difference between an intuitive statement about perpendicular lines and the point/ray representation used in the Hilbert layer.

1.8 Uniqueness of the Perpendicular Foot

The key geometric lemma is

```
theorem perpendicular_foot_unique
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P H K : Geo.Point)
  (hPerpH : PerpendicularToAxis Geo axis H P)
  (hPerpK : PerpendicularToAxis Geo axis K P) :
  H = K
```

The proof is neutral.

Assume $H \neq K$. Because both H and K lie on the reflection axis, carrier transport converts the two perpendicularity witnesses into right angles

$$\angle KHP \quad \text{and} \quad \angle HKP.$$

Hence the nondegenerate triangle HKP would have right angles at both H and K .

Extend KH beyond H . The resulting exterior angle is again right. All right angles are congruent, so this exterior angle is congruent to the remote interior right angle at K . Hilbert's exterior-angle theorem forbids this. Therefore $H = K$.

The argument uses no Euclidean parallel axiom. Its mathematical core is

two distinct perpendicular feet $\implies$ triangle with two right angles $\implies \perp$.

1.9 Uniqueness of Reflection

Once the perpendicular foot is unique, uniqueness of the reflected point follows from the existing segment-construction uniqueness theorem.

```

theorem line_reflection_unique
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P P1 P2 : Geo.Point)
  (hRef1 : IsLineReflection Geo axis P P1)
  (hRef2 : IsLineReflection Geo axis P P2) :
  P1 = P2

```

In the off-axis case, both reflections have the same perpendicular foot H . Both points P_1, P_2 lie on the prescribed ray beyond H , and their distances are determined by the same midpoint data. Segment additivity and `hilbert_segment_construction_unique` identify the two endpoints.

Thus

$$\forall P \exists! P' \text{ IsLineReflection}(l, P, P').$$

This is the logical threshold at which reflection can be promoted from a relation to a point transformation.

1.10 The Reflection Operator

The actual point transformation is selected noncomputably from the existence theorem:

```

noncomputable def lineReflect
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P : Geo.Point) :
  Geo.Point :=
  Classical.choose
  (line_reflection_exists Geo axis P)

```

Its specification theorem is

```

theorem lineReflect_spec
  ...
  (P : Geo.Point) :
  IsLineReflection Geo axis P
  (lineReflect Geo axis P)

```

The use of `Classical.choose` is a packaging step only. The geometric existence and uniqueness were proved before the function was introduced.

1.11 The First Coxeter Relation

The main result of the module is

```
theorem lineReflect_involutive
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo)
  (P : Geo.Point) :
  lineReflect Geo axis
    (lineReflect Geo axis P) = P
```

The proof has a short conceptual form:

$$\begin{array}{l}
 P \ R_l \ r_l(P) \\
 \Downarrow \text{ symmetry} \\
 r_l(P) \ R_l \ P \\
 r_l(P) \ R_l \ r_l(r_l(P)) \\
 \Downarrow \text{ uniqueness} \\
 P = r_l(r_l(P)).
 \end{array}$$

Consequently the first Coxeter relation has now been derived inside the Hilbert model:

$$\boxed{r_l^2 = 1.}$$

This is stronger architecturally than merely assuming an abstract involutory generator: the generator itself has been reconstructed from incidence, order, congruence, midpoint, and right-angle geometry.

1.12 Permutation Packaging

The reflection is finally packaged as a permutation of the point set:

```
noncomputable def lineReflectEquiv
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (axis : ReflectionAxis Geo) :
  Equiv Geo.Point Geo.Point
```

Both the forward and inverse maps are `lineReflect`; the two inverse laws are exactly `lineReflect_involutive`.

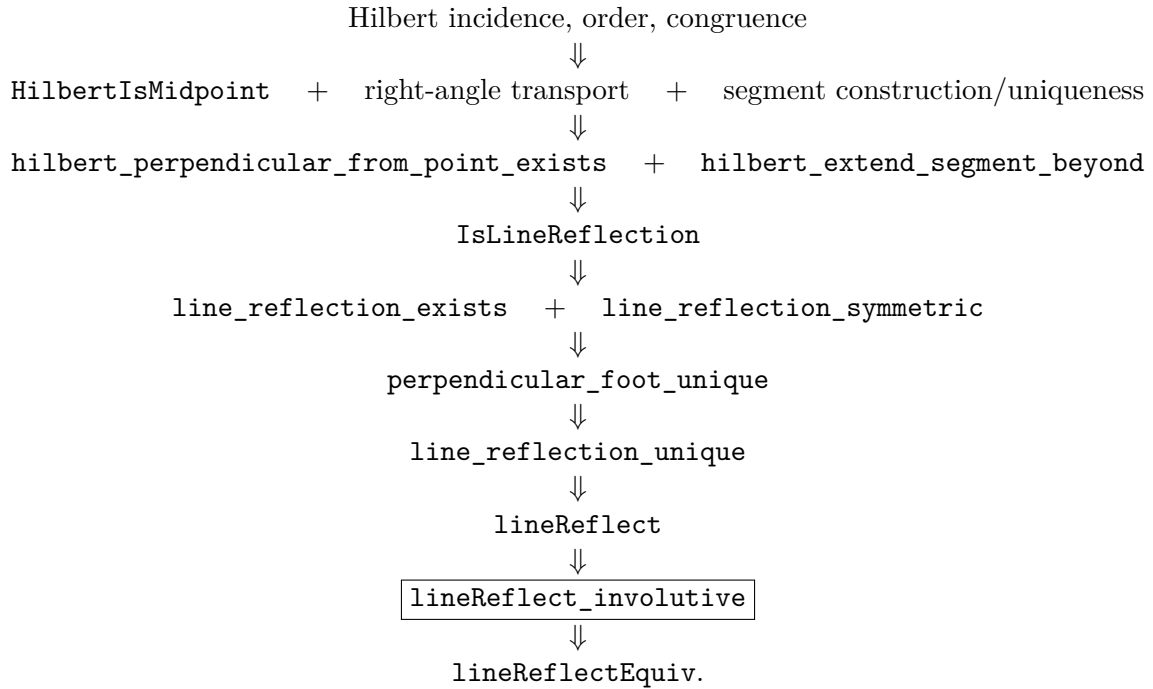
The associated simplification lemmas are

```
lineReflectEquiv_apply
lineReflectEquiv_apply_twice
```

This packaging is the interface needed by the next module, where different reflection generators can be composed into words.

1.13 Dependency Summary

The formal architecture may be summarized as follows:



The only direct Euclid Proposition import required by this module is the Hilbert reconstruction of I.12 for the perpendicular construction. The remaining work belongs to the established Hilbert congruence/order/right angle infrastructure and the local Coxeter helper layer.

1.14 Axiomatic Status

The module introduces no new declared geometric axiom and no **sorry**. The uniqueness of the perpendicular foot is proved by neutral angle theory rather than by uniqueness of parallels.

The transformation `lineReflect` is declared **noncomputable** because it uses `Classical.choose` after existence has been established. This should not be confused with a new geometric axiom.

The exact transitive axiom set should be recorded with `#print axioms` when the Coxeter checkpoint is audited. The present documentation therefore makes the precise local claim: *no new axiom is introduced by `Reflection.lean`.*

1.15 Architectural Significance

The development establishes a new path in CGJteamLab:

Hilbert geometry $\longrightarrow$ geometrically defined reflection $\longrightarrow$ point permutation $\longrightarrow$ group word.

This is distinct from two earlier proof mechanisms:

- a Pappus-style relabelling argument, where the same points are read in a different order;
- direct Hilbert congruence proofs, where no transformation object is constructed.

Here the transformation is a first-class object derived from the Hilbert geometry. This is the feature required before Coxeter words and pairwise-generator relations can be studied formally.

1.16 Next Checkpoint

No further material should be added to `Reflection.lean` merely to anticipate later needs.

The next experimental module is

```
CGJteamLab/Coxeter/ReflectionWords.lean
```

with the initial task of composing two generators

$$r_l r_m.$$

Only after this composition layer is stable should the project attempt the second Coxeter relation

$$(r_l r_m)^{p_{lm}} = 1$$

for intersecting axes, and the infinite-period case associated with parallel axes.

1.17 Conclusion

The first Coxeter generator has been reconstructed rather than postulated.

Starting from Hilbert incidence, order, congruence, midpoint, and right-angle geometry, the development proves existence and uniqueness of line reflection, promotes the resulting relation to a permutation of the point set, and establishes

$$\boxed{r_l^2 = 1}.$$

This provides the first formal bridge from the synthetic Hilbert language of CGJteamLab to a generators-and-relations language of geometric transformations.

Appendix A

Source Map: Coxeter–Moser

The supplement currently uses the following parts of Coxeter–Moser as conceptual control.

A.1 Section 1.1: Generators and Relations

This section supplies the abstract language of generators, words, relations, and presentations.

A.2 Section 4.1: Cyclic and Dihedral Groups

This section provides the elementary geometric interpretation of dihedral groups in terms of rotations and reflections.

A.3 Section 4.3: Groups Generated by Reflections

This section connects words in reflection generators with paths through fundamental regions and explains how local geometric circuits produce defining relations.

A.4 Sections 9.1–9.3: Groups Generated by Reflections

These sections provide the general Coxeter-type presentation

$$r_i^2 = 1, \quad (r_i r_j)^{p_{ij}} = 1,$$

the graphical notation, and the realization of generators by real affine reflections.

Appendix B

Lean Module Map

The intended module architecture is

```
CGJteamLab/Coxeter/  
  Reflection.lean  
  ReflectionWords.lean  
  CoxeterRelations.lean  
  EuclidBenchmarks.lean
```

At the current checkpoint only `Reflection.lean` is part of the completed Coxeter layer.

Checkpoint

Current completed milestone:

Hilbert line reflection $\implies$ permutation of points $\implies r_l^2 = 1$

The next planned module is `CGJteamLab/Coxeter/ReflectionWords.lean`.